

1 AssociationExplorer: A user-friendly Shiny application 2 for exploring associations and visual patterns

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10 Abstract

AssociationExplorer is an open-source interactive R Shiny application designed to help non-technical users explore statistical associations within multivariate datasets. Aimed particularly at journalists, educators, and engaged citizens, the tool facilitates the discovery and interpretation of meaningful patterns between variables without requiring programming or statistical expertise. Users can upload structured data (e.g., from surveys or open government datasets), select relevant variables, and dynamically visualize relationships via a correlation network and contextual bivariate plots. To illustrate its capabilities, we present a case study based on the European Social Survey (ESS), showcasing how users can investigate links between attitudes, behaviors, and socio-demographic indicators across countries. The app supports a range of association measures adapted to variable types (Pearson's r , Eta, and Cramer's V), ensuring both flexibility and statistical rigor. The visual interface enables users to adjust thresholds for association strength and examine results through interactive graphs and summary tables, making the app particularly well-suited for data storytelling, exploratory research, and public communication. AssociationExplorer demonstrates how open-source statistical tooling can enhance transparency, accessibility, and insight in the interpretation of complex social data.

11 **Keywords:** R Shiny, Exploratory data analysis, Correlation network

Nr.	Code metadata description	Metadata
C1	Current code version	v3.5.5
C2	Permanent link to code/repository used for this code version	https://github.com/AntoineSoetewey/AssociationExplorer
C3	Permanent link to Reproducible Capsule	https://codeocean.com/capsule/7707308/tree
C4	Legal Code License	MIT License
C5	Code versioning system used	Git/GitHub
C6	Software code languages, tools, and services used	R, R Shiny
C7	Compilation requirements, operating environments & dependencies	See a list of the required R packages at https://github.com/AntoineSoetewey/AssociationExplorer/blob/main/packages.md
C8	If available link to developer documentation/manual/screencast	https://github.com/AntoineSoetewey/AssociationExplorer/tree/main/documentation
C9	Support for questions or issues	https://github.com/AntoineSoetewey/AssociationExplorer/issues

Table 1: Code metadata

12 Metadata

13 The metadata associated with the current version of the software is summa-
14 rized in Table 1.

15 The remainder of this paper is structured as follows. Section 1 outlines
16 the motivation and significance of the application, highlighting its contribu-
17 tion to data accessibility and exploratory analysis. Section 2 provides a de-
18 tailed description of the software’s architecture and functionalities. Section 3
19 presents an illustrative example using data from the European Social Sur-
20 vey. Section 4 discusses the potential impact of the application in research,

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21 education, and journalism. Finally, Section 5 concludes with a summary and
22 future directions.

23 1. Motivation and significance

24 The growing availability of large, complex, and high-dimensional datasets in
25 the social sciences and public policy domains offers unprecedented opportu-
26 nities for insight but also presents significant challenges for exploration and
27 interpretation, particularly for non-specialist audiences. Journalists, educa-
28 tors, and engaged citizens often struggle to identify and interpret meaningful
29 relationships between variables without the aid of programming skills or for-
30 mal statistical training. This barrier limits the broader societal impact of
31 open data initiatives, which are designed to promote transparency, account-
32 ability, and informed public discourse.

33 To address this gap, we developed AssociationExplorer, a free, open-source
34 R Shiny [1] application that enables intuitive and statistically grounded ex-
35 ploration of multivariate associations. The tool guides users through a visual
36 journey of variable relationships by automatically computing appropriate bi-
37 variate association measures—Pearson’s r , Eta, and Cramer’s V —depending
38 on variable types, and presenting the results in an interactive correlation
39 network. Users can set thresholds for the strength of association and explore
40 linked bivariate plots or tables with descriptive labels. This workflow sup-
41 ports transparent, reproducible, and non-technical exploratory data analysis
42 (EDA).

43 Our software is particularly suited to survey-based datasets and public opin-
44 ion studies. As an illustrative case, we apply AssociationExplorer to the
45 European Social Survey (ESS), a cross-national survey that collects attitudi-
46 nal, behavioral, and socio-demographic data across European countries. The
47 tool allows users to uncover associations between trust in institutions, pol-
48 icy preferences, media usage, and demographic characteristics without any
49 coding. This type of interactive analysis can empower journalists to build
50 data-driven narratives, educators to teach statistical thinking, and citizens
51 to explore evidence underlying public debate.

52 While several tools and libraries exist for correlation analysis (e.g., `corrr`
53 [8], `GGally` [18], `corrplot` [22], `ggstatsplot` [14], `correlation` [11, 12],
54 `lares` [9] and `Hmisc` [7] in R [15], or Python packages like `seaborn` [21]
55 and `pingouin` [20]), they typically require programming proficiency and fo-
56 cus primarily on numerical associations. Most of these tools do not handle
57 nominal categorical variables directly; if included, such variables are often
58 transformed using one-hot or dummy encoding, which can transform their
59 original structure and limit interpretations.

60 In contrast, AssociationExplorer is designed to handle both quantitative and
61 qualitative variables (including nominal factors) natively and transparently.
62 It provides a guided, end-to-end workflow that begins with data upload and
63 preprocessing, continues through variable selection and association filtering,
64 and ends with interpretable visualizations. This structured process is in-
65 tuitive and accessible for users of all backgrounds, making the app espe-
66 cially suitable for those without programming experience or formal statistical
67 training. By lowering the technical barrier for statistical exploration, Asso-
68 ciationExplorer contributes to a more inclusive data culture and supports
69 data-driven discovery in both academic and public-facing contexts.

70 2. Software description

71 2.1. Software architecture

72 Upon upload, the dataset is preprocessed to exclude variables with zero vari-
73 ance, as these variables do not vary across observations and therefore cannot
74 contribute to meaningful associations or visualizations. Removing them helps
75 reduce noise and ensures that only informative variables are included in the
76 analysis. Optionally, the user can provide a variable description file, which
77 is integrated and used to annotate visual elements. The backend computes
78 association measures tailored to the variable types: Pearson's r for numeric
79 pairs, Cramer's V for categorical pairs, and the correlation ratio (η) for mixed
80 pairs. Associations are filtered using user-defined thresholds and represented
81 in a correlation network and complementary bivariate plots. The app handles
82 both CSV and Excel files.

83 The application is currently available as a standalone open-source R Shiny
84 application on GitHub.¹ It can be launched directly from an R session using
85 the following command, provided the `shiny` package is installed:

```
86 library(shiny)  
87 runGitHub("AssociationExplorer", "AntoineSoetewey")
```

88 A fully web-based version, requiring no installation or technical setup, is also
89 planned as part of its integration into a broader online platform developed
90 within the ODALON (Open multimodal Data for Automated Local News)
91 research project, which aims to support the production of local news (more
92 information about this project in Sections 3 and 4).

¹<https://github.com/AntoineSoetewey/AssociationExplorer>

93 *2.2. Software functionalities*

94 The major functionalities of the AssociationExplorer application include:

- 95 • **Data upload and cleaning:** The app supports CSV and Excel files.
 96 It automatically removes variables with only one unique value, as they
 97 lack variability and cannot contribute to association analyses. Addi-
 98 tionally, it can optionally integrate user-supplied descriptions of vari-
 99 ables, which are used to enhance the clarity and interpretability of
 100 visualizations, particularly for non-technical users. The description file
 101 must be structured with two columns: **Variable** (containing the ex-
 102 act variable names as in the dataset) and **Description** (containing the
 103 corresponding descriptions).
- 104 • **Variable selection interface:** Users can interactively choose which
 105 variables to explore. When a description file is provided, a summary
 106 table links variable names to their descriptions.
- 107 • **Dynamic association filtering:** The app computes pairwise associ-
 108 ation measures between all selected variables, using a method tailored
 109 to the types of variables involved:
 - 110 – For pairs of numeric variables X and Y , the app calculates Pear-
 111 son’s correlation coefficient (r), and retains the association if the
 112 coefficient of determination (R^2) exceeds a user-defined threshold:

$$R^2 = r^2 = (\text{cor}(X, Y))^2 \quad (1)$$

113 where the Pearson’s correlation coefficient $\text{cor}(X, Y)$ is defined as:

$$r(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (2)$$

114 where $\bar{X}$ and $\bar{Y}$ are the sample means of X and Y , respectively,
 115 and n is the number of observations.

- 116 – For pairs of categorical variables, it computes Cramer’s V , a nor-
 117 malized measure of association derived from the chi-squared statis-
 118 tic:

$$V = \sqrt{\frac{\chi^2}{n \cdot \min(k - 1, r - 1)}} \quad (3)$$

119 where χ^2 is the chi-squared statistic, n is the total number of ob-
 120 servations, and k, r are the numbers of categories in each variable.

121 – For mixed pairs (one numeric and one categorical variable), the
 122 app computes the correlation ratio (η), which quantifies how much
 123 of the variance in the numeric variable is explained by the grouping
 124 structure of the categorical variable. It is defined as:

$$\eta = \sqrt{\frac{SS_{\text{between}}}{SS_{\text{total}}}} \quad (4)$$

125 where:

126 - SS_{total} is the *total sum of squares* of the numeric variable:

$$SS_{\text{total}} = \sum_{i=1}^n (y_i - \bar{y})^2$$

127 with y_i the observed numeric values and $\bar{y}$ their overall mean.

128 - SS_{between} is the *between-group sum of squares*, computed as:

$$SS_{\text{between}} = \sum_{g=1}^G n_g (\bar{y}_g - \bar{y})^2$$

129 where G is the number of groups (categories), n_g is the number
 130 of observations in group g , $\bar{y}_g$ is the group mean, and $\bar{y}$ is the
 131 overall mean.

132 This formulation captures the proportion of the total variance in
 133 the numeric variable that can be attributed to differences between
 134 the categorical groups. A pair is retained only if η^2 exceeds the
 135 numeric threshold defined by the user.

136 Each association is retained only if its corresponding strength metric—
 137 R^2 , η^2 , or Cramer’s V —exceeds the threshold set by the user. These
 138 thresholds can be adjusted interactively through the interface, and
 139 the filtering process is reactive: updates to the thresholds immediately
 140 propagate to the network and bivariate visualizations. This allows users
 141 to dynamically control the sensitivity of the association analysis and
 142 focus on relationships of substantive interest.

143 • **Interactive correlation network:** The filtered associations are dis-
 144 played as an interactive graph where nodes represent variables and
 145 edges represent associations. Edge thickness and length reflect the
 146 strength of the association: stronger associations are shown with thicker

and shorter edges, whereas weaker associations are displayed with thinner and longer edges. For quantitative–quantitative pairs specifically, the color of the edges conveys direction: red edges indicate negative associations, while blue edges indicate positive ones. This combination of visual cues helps users quickly identify the most meaningful relationships in the network. When hovering over nodes, variable descriptions are displayed if a description file has been provided by the user; otherwise, the variable names are shown by default. This allows for quick access to additional context without cluttering the visualization, enhancing interpretability while keeping the network clean and readable. The network is built using the `visNetwork` R package, which supports interactive and customizable graph layouts. Full documentation is available at <https://datastorm-open.github.io/visNetwork/>.

- **Bivariate visualization of variable pairs:** For each variable pair exceeding the threshold:

- Scatter plots with linear regression lines are shown for numeric pairs, helping visualize the direction and strength of the relationship. The regression line is computed using the ordinary least squares (OLS) method, which minimizes the sum of squared vertical distances between the observed data points and the fitted line.
- Colored contingency tables with marginal sums are shown for categorical pairs, where cell background colors vary in intensity according to the frequency of observations, using a blue gradient to highlight higher counts.
- Mean plots are shown for numeric-categorical pairs, with bars ordered by mean value to make it easy to compare and rank categories based on the quantitative variable.

Confidence intervals for the regression lines and standard errors in the mean plots are intentionally omitted to maintain a clean, uncluttered visualization that prioritizes ease of interpretation. Mean plots were selected over boxplots to avoid overwhelming non-expert users with distributional information, focusing instead on clear, accessible insights about average group differences.

- **Accessibility and user guidance:** A dedicated help section explains each step, allowing users with a limited statistical background to interactively explore their data.

184 *2.3. Sample code snippets analysis*

185 Below is a representative snippet from the application showing how the soft-
186 ware selects the appropriate association measure depending on the types of
187 the variable pair and filters associations based on user thresholds:

```
188 # Numeric vs numeric case
189 if (is_num1 && is_num2) {
190     ...
191     r <- cor(x, y, use = "complete.obs")
192     cor_val <- ifelse(r^2 >= threshold_num, r, 0)
193     cor_type <- "Pearson's r"
194
195 # Categorical vs categorical case
196 } else if (!is_num1 && !is_num2) {
197     ...
198     tbl <- table(x, y)
199     ...
200     n_obs <- sum(tbl)
201     df_min <- min(nrow(tbl) - 1, ncol(tbl) - 1)
202     if (df_min > 0) {
203         v_cramer <- sqrt(chi$statistic / (n_obs * df_min))
204         cor_val <- ifelse(v_cramer >= threshold_cat,
205                           v_cramer, 0)
206         cor_type <- "Cramer's V"
207     }
208
209 # Mixed case (numeric vs categorical)
210 } else {
211     ...
212     means_by_group <- tapply(num_var, cat_var,
213                               mean, na.rm = TRUE)
214     overall_mean <- mean(num_var, na.rm = TRUE)
215     n_groups <- tapply(num_var, cat_var, length)
216     bss <- sum(n_groups * (means_by_group - overall_mean)^2,
217               na.rm = TRUE)
218     tss <- sum((num_var - overall_mean)^2, na.rm = TRUE)
219
220     if (tss > 0) {
221         eta <- sqrt(bss / tss)
222         cor_val <- ifelse(eta^2 >= threshold_num, eta, 0)
223         cor_type <- "Eta"
```



```
224     }  
225 }
```

226 This conditional structure ensures that the correct statistical method is ap-
227 plied for each type of variable pair, supporting a robust and interpretable
228 exploration of associations.

229 3. Illustrative example

230 To demonstrate the core functionalities of AssociationExplorer, we use a cu-
231 rated subset of data from the European Social Survey (ESS), Round 11.
232 The ESS is a large-scale, cross-national survey that measures attitudes, be-
233 liefs, and behaviors across European countries. The original dataset includes
234 responses from over 46,000 individuals on topics such as politics, trust, well-
235 being, media use, and health. The full ESS dataset, codebook, and docu-
236 mentation are freely available at <https://ess.sikt.no/en/> [3, 2].

237 For this example, we focus on the Belgian respondents, resulting in a re-
238 duced dataset of 1,594 individuals. We selected 60 variables covering areas
239 highly relevant for understanding public opinion and everyday life in Belgium:
240 interest in politics, confidence in institutions, lifestyle behaviors, perceived
241 discrimination, vaccination, and more. These variables include both numbers
242 (quantitative data) and labels or categories (qualitative data), making the
243 dataset ideal for exploring diverse forms of associations.

244 This example is particularly relevant for our research project ODALON,
245 which aims to develop a platform that supports the (semi-)automated pro-
246 duction of local news in Belgium. AssociationExplorer plays a key role in
247 this effort by offering journalists, researchers, and citizens an intuitive tool to
248 explore potentially newsworthy patterns in public and survey data, without
249 requiring programming skills or statistical training.

250 Data preparation for the curated dataset was carried out in R and included:

- 251 • Filtering the dataset to include only Belgian respondents.
- 252 • Converting survey-specific nonresponse codes (e.g., 77, 88, 9999, etc.)
253 to NA values, based on the ESS codebook.
- 254 • Recoding several categorical variables to have meaningful and inter-
255 pretable labels (e.g., for gender, religion, political participation, or
256 health behaviors).

257 The full R script used to perform this transformation is openly available in the
258 data folder of the GitHub repository at [https://github.com/AntoineSoetewey/](https://github.com/AntoineSoetewey/AssociationExplorer/tree/main/shiny_app/data)
259 AssociationExplorer/tree/main/shiny_app/data.

Association Explorer

Upload your dataset (CSV or Excel)

Browse... ESS11_BE_data.csv
Upload complete

(Optional) Upload variable descriptions (CSV or Excel)

Browse... ESS11_BE_description.csv
Upload complete

The descriptions file must contain exactly two columns named 'Variable' and 'Description'.

Process data

Figure 1: Data upload tab. Users upload their dataset and can optionally provide a variable description file containing variable names and descriptions.

260 Once the dataset is uploaded into AssociationExplorer via the **Data** tab (see
261 Figure 1), users are guided through a step-by-step process. In addition to
262 the main dataset, users can optionally upload a separate description file that
263 provides human-readable explanations for each variable. In our example, this
264 file was created using information from the official ESS codebook, allowing
265 for clearer interpretation throughout the app interface. If no description
266 file is provided, the application will automatically use the variable names
267 themselves as default labels in all visualizations and summary tables.

268 In the **Variables** tab, they select the variables they wish to explore, op-
269 tionally assisted by a table of the variables' names and descriptions (see
270 Figure 2).

271 Next, users can adjust the association thresholds in the **Correlation Network**
272 tab to focus on the most meaningful relationships among the selected vari-
273 ables (see Figure 3).

274 Finally, the **Pairs Plots** tab displays detailed bivariate visualizations, in-
275 cluding scatter plots, mean plots, and contingency tables, for each retained
276 association (see Figures 4-6).

277 In addition, the **Help** tab provides a concise, step-by-step guide on how to
278 use the application (see Figure 7).

279 This example shows how non-expert users, such as journalists or engaged
280 citizens, can uncover unexpected or important relationships in a public opin-
281 ion dataset. These insights can serve as the starting point for local news,
282 public debates, or policy communication. A screencast demonstrating the

Association Explorer

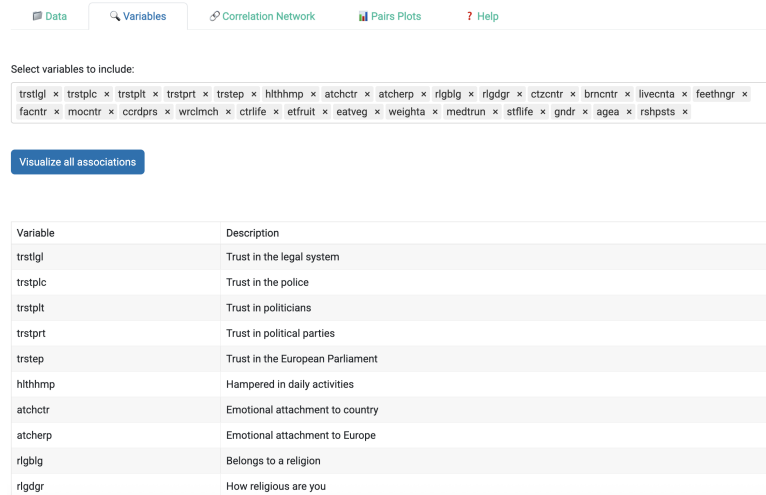


Figure 2: Variables tab. Users select the variables to include in the analysis, with access to descriptions (if provided in the data tab) for easier selection.

Association Explorer

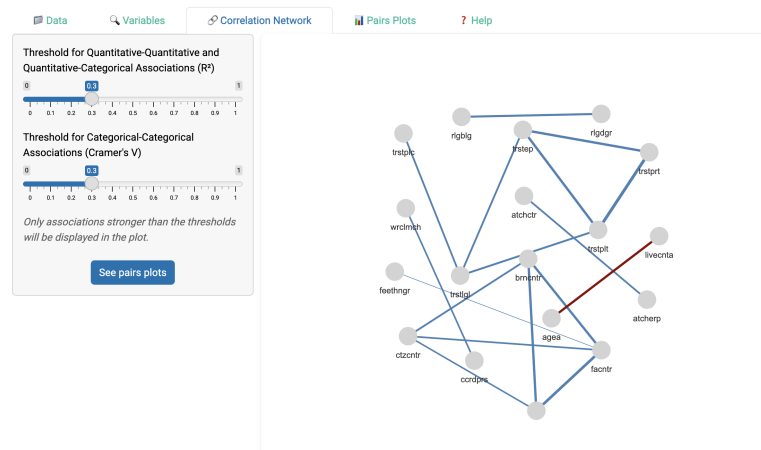


Figure 3: Interactive correlation network. Nodes represent variables and edges represent associations; edge thickness and length indicate strength, and edge color reflects direction (for numeric pairs only).

Association Explorer

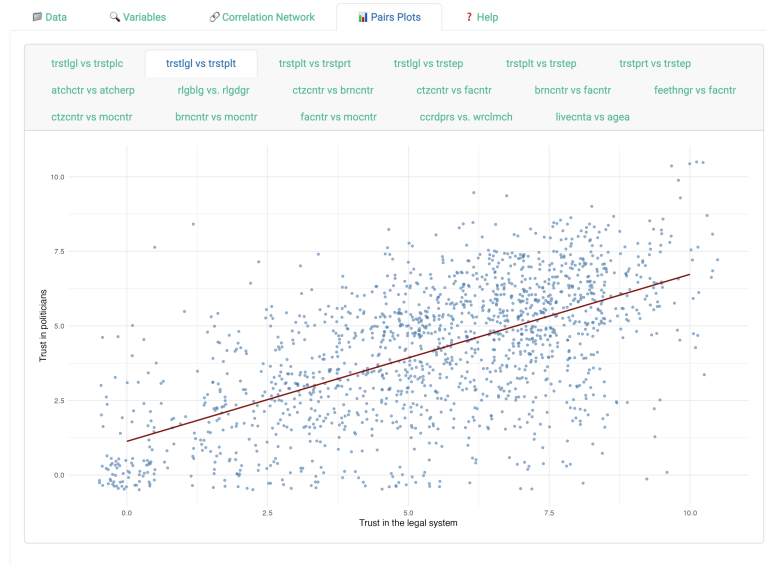


Figure 4: Example of a scatter plot for a numeric-numeric pair. The plot includes a linear regression line fitted via ordinary least squares (OLS).

Association Explorer



Figure 5: Example of a mean plot for a numeric-categorical pair. Categories are ordered by the mean value of the quantitative variable to aid interpretation.

Association Explorer

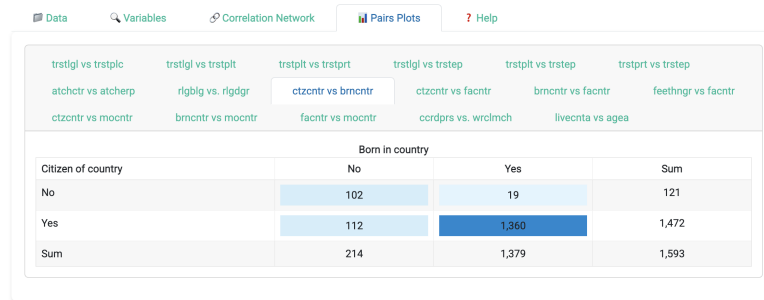


Figure 6: Example of a colored contingency table for a categorical-categorical pair, with color intensity reflecting frequencies and marginal sums displayed.

Association Explorer



How to use the Association Explorer app?

- Upload your dataset (CSV or Excel) in the 'Data' tab. Optionally, upload a file with variable descriptions. This file must contain 2 columns called 'Variable' and 'Description'.
- In the 'Variables' tab, select the variables you want to explore. If you upload a file containing variables' descriptions, a summary table below shows the selected variables along with their descriptions.
- Click 'Visualize all associations' to access the correlation network.
- Adjust the thresholds to filter associations by strength. Only variables that have strong associations (as defined by the thresholds) will appear in the network and pairs plots.
- In the network, thicker and shorter edges indicate stronger associations.
- Click 'See pairs plots' to display bivariate visualizations for retained associations.

Figure 7: Help tab, offering guidance on how to use the application.

283 full workflow of the application is available in the documentation folder of
284 the GitHub repository.

285 Note that in this example, no survey weights have been applied. While the
286 ESS dataset includes weights to account for complex sampling designs, we
287 chose not to use them here in order to maintain clarity and responsiveness
288 in the interface, and because our focus is on demonstrating the app’s core
289 functionality rather than producing nationally representative statistics.

290 4. Impact

291 AssociationExplorer enables a broader range of users, including researchers,
292 students, journalists, and engaged citizens, to explore complex datasets in-
293 volving both quantitative and qualitative variables. By eliminating the need
294 for advanced programming or statistical knowledge, the software lowers the
295 barrier to entry for data-driven discovery and interpretation. The app facil-
296 itates the systematic screening of pairwise associations in survey or public
297 data, offering insights that can guide hypothesis generation, variable selec-
298 tion, or further multivariate modeling.

299 It also facilitates the assessment of relationships between variables of dif-
300 ferent types (numeric, ordinal, and nominal) within a single unified inter-
301 face. Unlike many tools that require preprocessing steps such as dummy
302 coding, AssociationExplorer handles categorical variables natively and auto-
303 matically selects the appropriate statistical association measure (i.e., Pear-
304 son’s r , Cramer’s V , correlation ratio η). This ensures that users can obtain
305 meaningful and interpretable results without needing to master complex pre-
306 processing steps or statistical diagnostics.

307 Beyond supporting research, we are convinced that automating these statisti-
308 cal processes can contribute to the promotion of open public data repositories.
309 In fact, the scientific literature points to a chronic underuse of open public
310 data [16], particularly due to the resources and knowledge required to pro-
311 cess these data. By lowering this barrier to entry, this tool makes it possible
312 to utilize databases that would otherwise remain inactive. The exploitation
313 and understanding of these data responds to crucial issues of transparency
314 in public action [19].

315 Although this tool is open to anyone, its development was particularly in-
316 spired by its use in a journalistic context. Data journalism, in particular, is
317 one of those examples of virtuous practice that we believe is best suited to
318 supporting the open data movement. This practice, which relies on the use
319 of digital data as material for journalistic investigation [6], is now used to
320 analyze election results, explain climate change, and track the evolution of

321 health crises. However, it comes at a cost, which explains why this practice
 322 is still limited to a few large national media outlets.
 323 However, it would be detrimental to leave these practices inaccessible to local
 324 newsrooms. By covering events that are often overlooked by national media,
 325 these local media outlets can highlight data that would otherwise be ignored
 326 (finance, consumption, police, municipal council records, etc.). Supporting
 327 data journalism at the local level can also help to enhance the value of this
 328 journalistic practice, which has been shown to strengthen social cohesion [13],
 329 stimulate civic participation [17], and influence public policies [10].
 330 While national media outlets are struggling, these challenges are even more
 331 pronounced for local media, which generally face the same problems but
 332 with even more limited resources [5]. As a result, few institutions have the
 333 means to meet the labor and skill requirements necessary for this investigative
 334 practice [4]. We hope that AssociationExplorer can help meet these needs.
 335 The app is also planned to be used as an interactive teaching aid in statistics
 336 and data literacy courses. It will allow students to upload real datasets and
 337 immediately visualize how variables relate, reinforcing theoretical concepts
 338 through intuitive, hands-on exploration. By presenting associations visually,
 339 the app encourages learners to engage with their data more actively and
 340 reflect critically on patterns, variable relationships, and data quality.
 341 Although already fully functional, the software is currently made available as
 342 a standalone open-source R Shiny application on GitHub and can be launched
 343 directly from R. At this stage, it is therefore only accessible to users who have
 344 R installed on their machine. As part of a broader research project called
 345 ODALON, which aims to build an integrated platform combining multiple
 346 tools for automated local news production, AssociationExplorer will serve
 347 as a key building block. A public, fully online version of the app will be
 348 released as the project progresses, and no later than December 2026 (which
 349 corresponds to the end of the project), ensuring that anyone, including those
 350 without access to R, can freely use the application through a user-friendly
 351 web interface.
 352 Future impact is expected through integration into platforms for civic data
 353 storytelling, local journalism, and public communication. While not intended
 354 for commercialization, the application's role in supporting open, inclusive,
 355 and responsible data exploration reflects its potential for broader societal
 356 relevance.

357 **5. Conclusions**

358 AssociationExplorer provides a user-friendly, open-source solution for explor-
 359 ing statistical associations within multivariate datasets. By integrating ro-

360 bust statistical measures with dynamic and intuitive visualizations, the app
361 empowers non-expert users, such as journalists, educators, and engaged cit-
362 izens, to uncover patterns and relationships that might otherwise remain
363 hidden. Its ability to handle both quantitative and qualitative variables,
364 combined with an interactive interface and an easy-to-follow workflow, makes
365 it especially suitable for exploratory research, data storytelling, and public
366 communication.

367 Beyond its technical features, the application contributes to democratizing
368 access to statistical tools and supports data literacy in a wide range of con-
369 texts. The case study using the European Social Survey illustrates how users
370 can navigate from raw data to meaningful insights with minimal technical
371 knowledge.

372 While the app is currently accessible through R for users with basic technical
373 setup, a fully web-based version will make it universally available as part of
374 a broader data exploration platform being developed under the ODALON
375 project. By lowering barriers to data exploration and interpretation, Asso-
376 ciationExplorer supports more inclusive, transparent, and evidence-informed
377 engagement with complex social data.

378 Future improvements will be guided by user feedback and community con-
379 tributions. One planned feature is the ability to incorporate user-specified
380 weights, especially relevant for survey data. However, this functionality is
381 not yet implemented in the current version to preserve simplicity and ensure
382 responsiveness in the user interface, particularly for users unfamiliar with
383 survey weighting procedures.

384 Additional avenues for future research and development include extending
385 the application to compute partial correlations, which can help isolate direct
386 relationships between variables by controlling for confounders. Visualizations
387 involving three variables could also be introduced, for instance, scatter plots
388 where the size or color of points reflects a third variable, to uncover more
389 nuanced patterns. Furthermore, users could be allowed to filter variables
390 based on missingness thresholds or variable types, and a module for longi-
391 tudinal data (e.g., panel structures or time series) could expand the app's
392 relevance to temporal analyses. For contingency tables, adding conditional
393 distributions (by row or by column) in addition to the joint distribution
394 could provide more informative insights. This would allow users to better
395 explore subsets of the data. For example, even if a group such is underrep-
396 resented in the sample (and therefore lightly colored in the joint table), the
397 conditional distribution would still reveal the relative patterns within that
398 group. Another enhancement could involve enabling two-sided intervals for
399 the association thresholds, rather than the current one-sided sliders. This
400 would allow users not only to focus on the strongest associations, but also to

deliberately examine the weakest ones, thereby uncovering cases where variables appear largely independent. In addition, providing interpretive text alongside tables and plots (especially if the application is deployed for non-specialist audiences) could help ensure that outputs are understood correctly and used responsibly. Lastly, adding options for exporting results or generating automated summaries of key associations could further enhance its use in reporting and collaborative workflows.

A more ambitious extension would integrate a large language model (LLM) to enable users to enter a free-text prompt or a thematic query. The app would then automatically identify variables that are semantically similar or potentially associated with the theme, based on the dataset and the optional description file provided by the user. These extracted variables would be pre-selected in the interface, as if the user had chosen them manually, after which the analysis would proceed as in the current version. This feature could significantly enhance accessibility for users unfamiliar with the dataset or its structure, and accelerate the discovery of relevant associations.

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